

Business Intelligence Report

Generated by DataSage AI

Generated On: 24 September 2026

1. Dataset Overview

DATASET OVERVIEW

Rows: 100
Columns: 10
Missing Values: 0
Duplicate Rows: 0

DESCRIPTIVE STATISTICS

Units_Sold
Average: 5128.71
Minimum: 124.00
Maximum: 9925.00

Unit_Price
Average: 276.76
Minimum: 9.33
Maximum: 668.27

KPI Columns:
- Units_Sold
- Unit_Price

Category Columns:
- Region
- Country
- Item_Type
- Sales_Channel
- Order_Priority

Date Column: Ship_Date

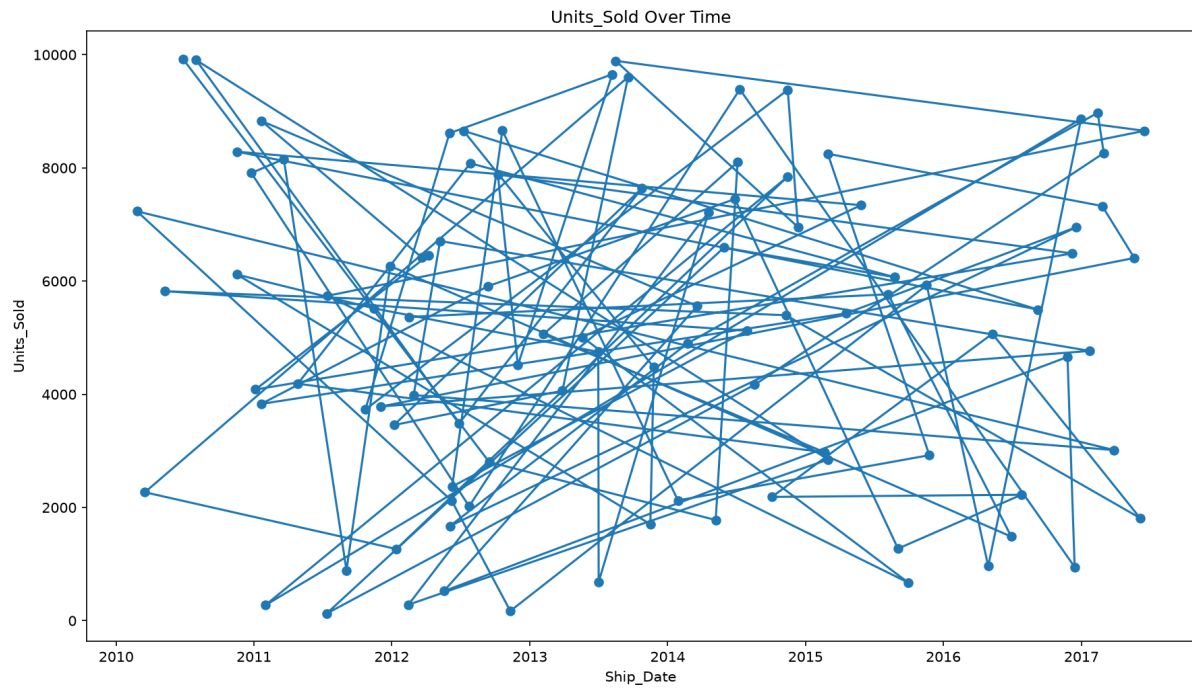
Most Common Region: Sub-Saharan Africa
Most Common Country: The Gambia

CORRELATION ANALYSIS

Strongest Negative: Order_ID ↔ Units_Sold (-0.22)

2. Visual Analytics

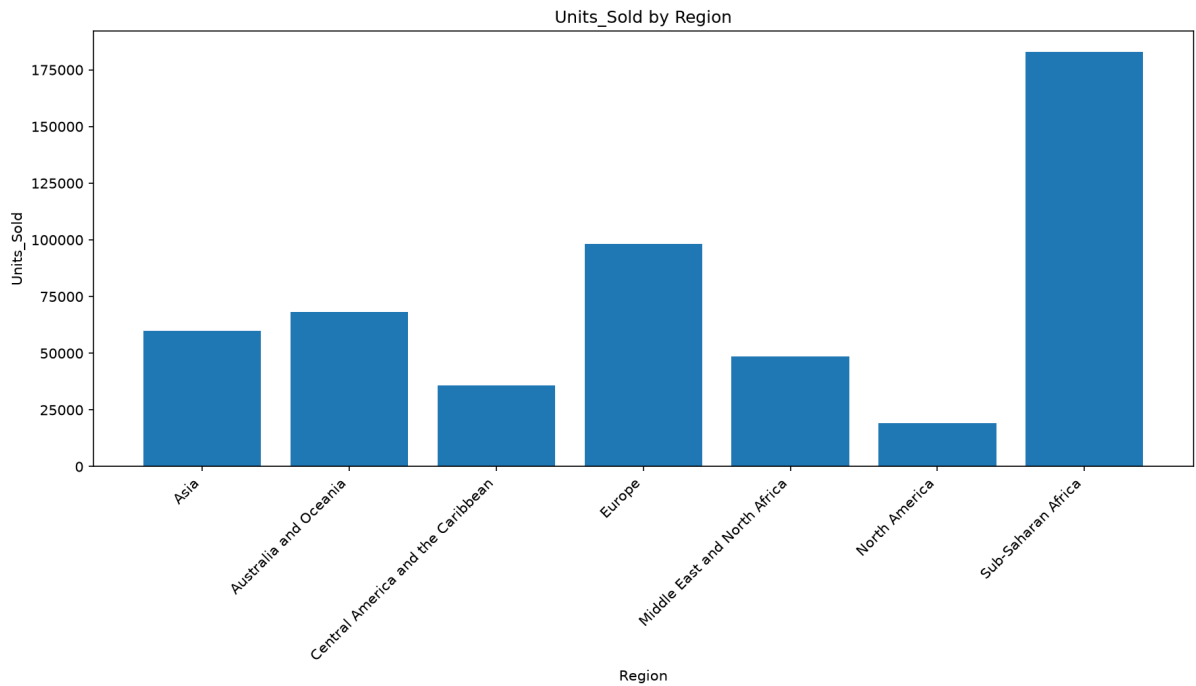
■ **Units_Sold Over Time**



■ **KEY INSIGHT**

- ■ Trend Analysis The Units_Sold has decreased over time.
- Start: 9925.
- 00 End: 5367.
- 00.

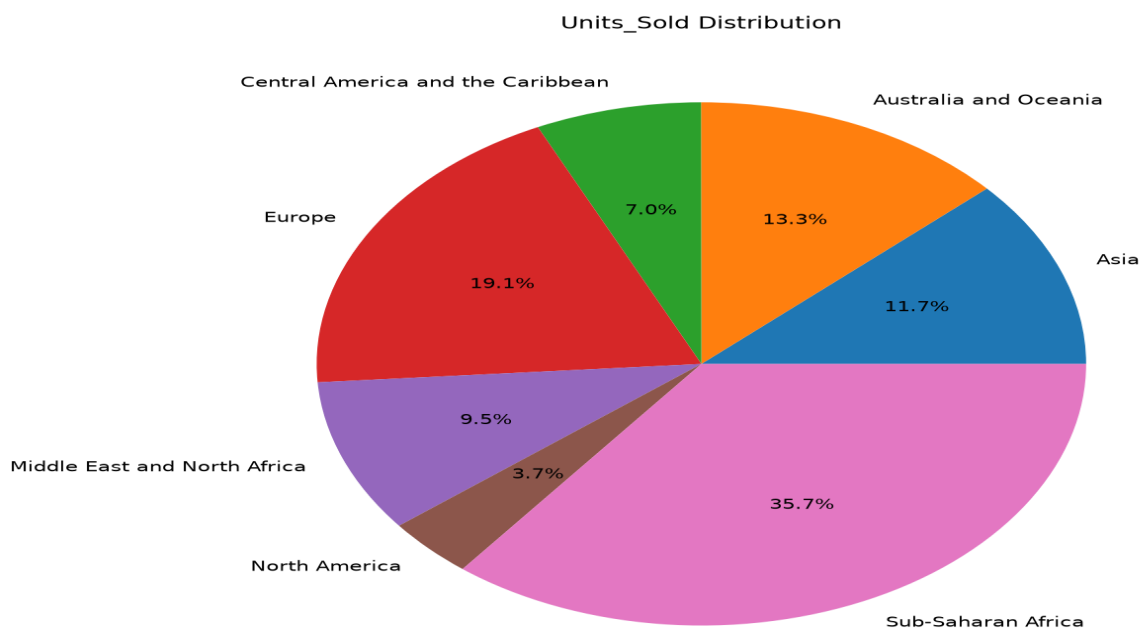
■ **Units_Sold by Region**



■ KEY INSIGHT

- ■ Category Analysis ■ Highest Category: Sub-Saharan Africa Value: 182870.
- 00 ■ Lowest Category: North America Value: 19143.
- 00.

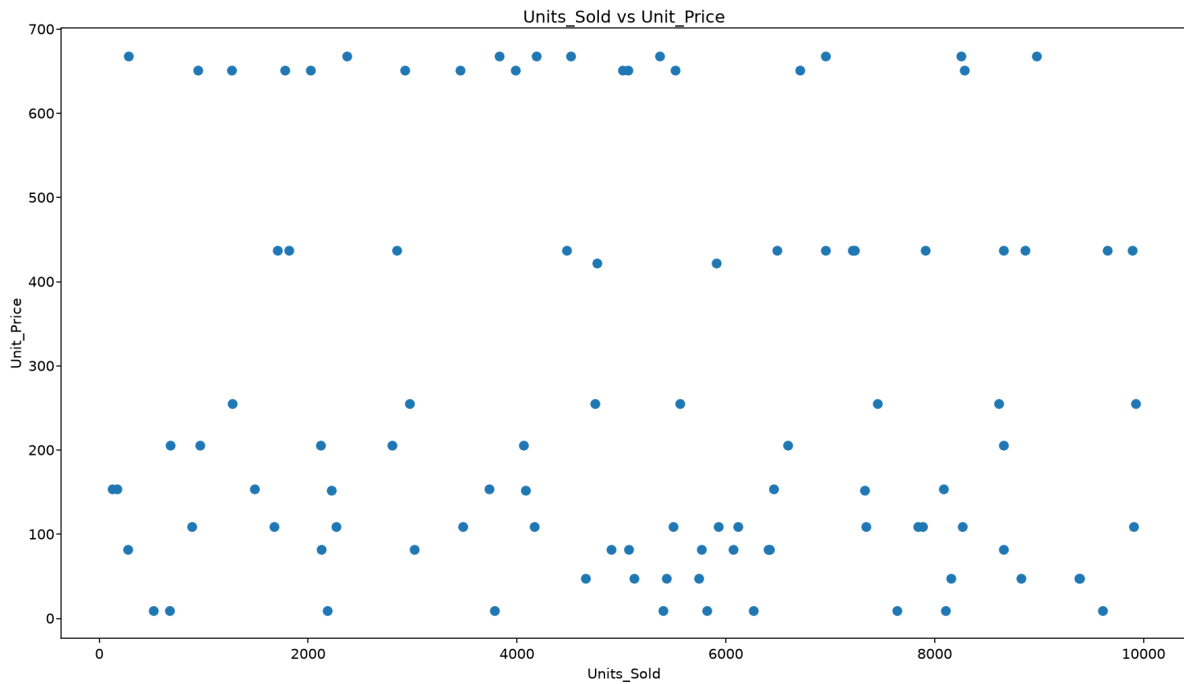
■ Units_Sold Distribution



■ KEY INSIGHT

- ■ Distribution Insight Sub-Saharan Africa contributes 35.
- 7% of the total.

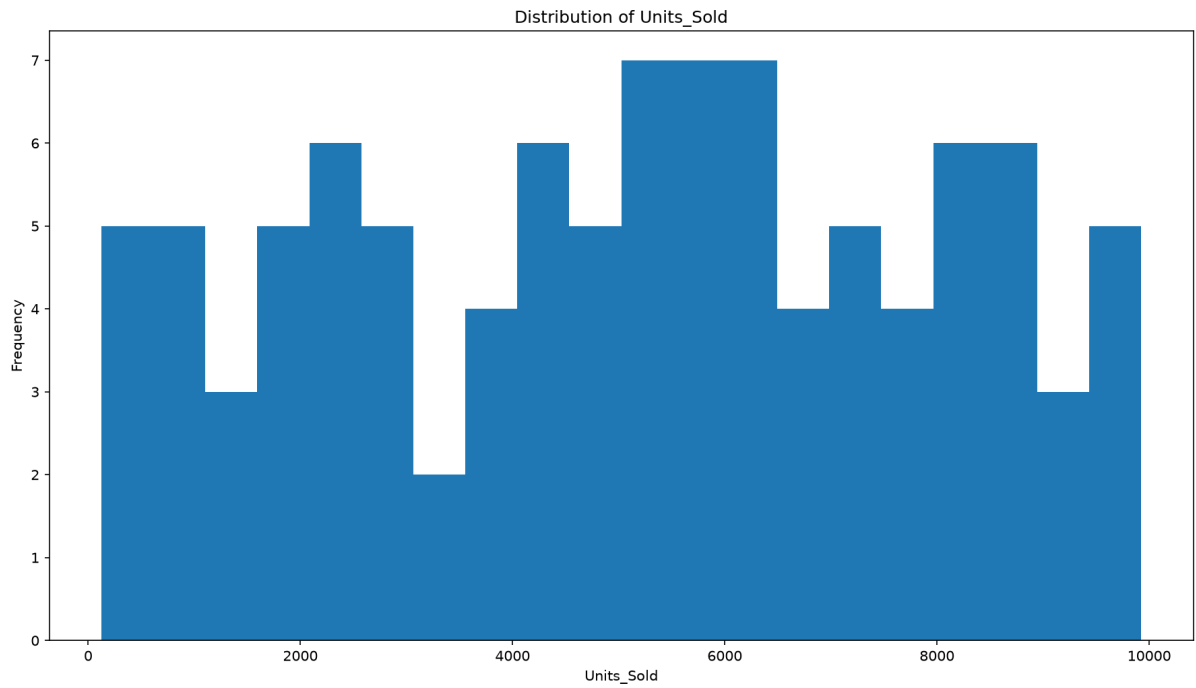
■ *Units_Sold vs Unit_Price*



■ KEY INSIGHT

- ■ Relationship Analysis There is a weak relationship between Units_Sold and Unit_Price.
- Correlation = -0.
- 07.

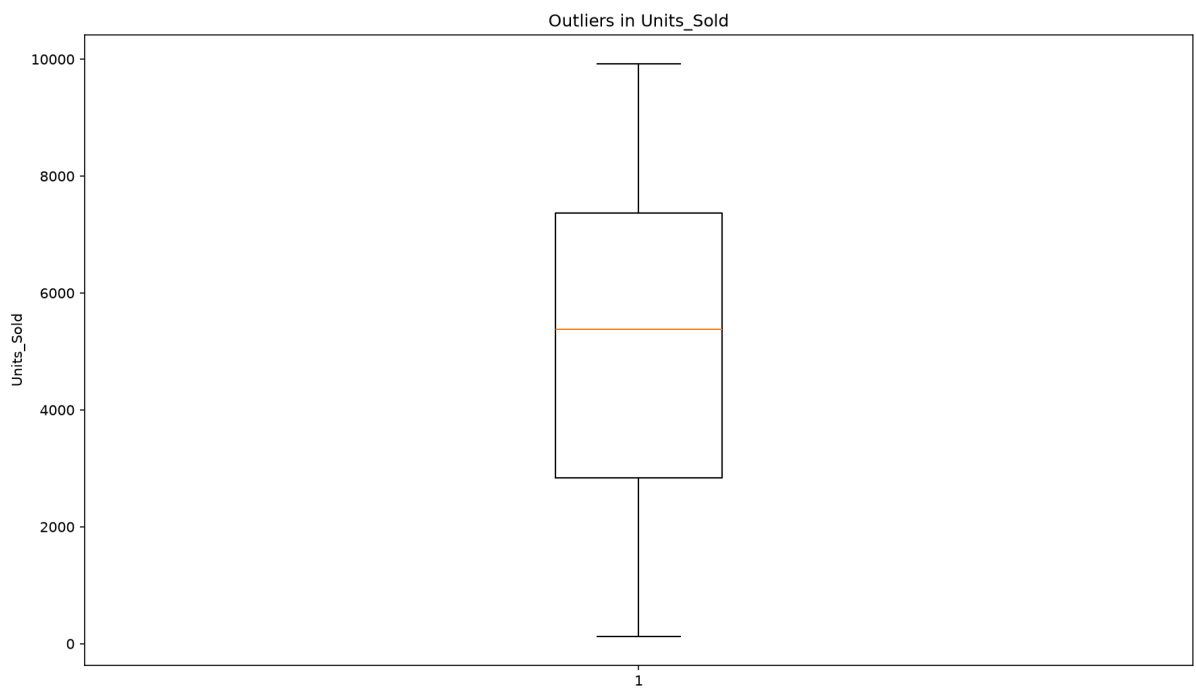
■ *Distribution of Units_Sold*



■ KEY INSIGHT

- ■ Distribution Analysis Average: 5128.
- 71 Median: 5382.
- 50 The data appears left-skewed.

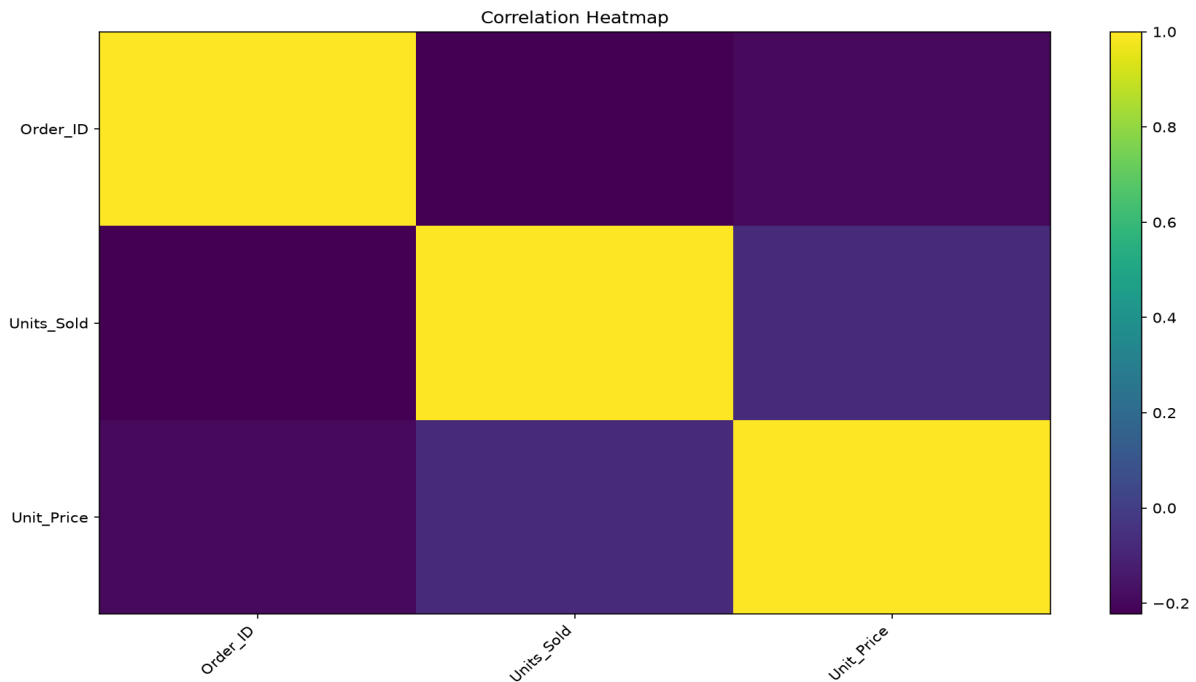
■ Outliers in Units_Sold



■ KEY INSIGHT

- ■ Outlier Analysis Detected 0 potential outliers.

■ Correlation Heatmap



■ KEY INSIGHT

- ■ Correlation Analysis The heatmap highlights positive and negative relationships between numeric variables.

3. AI Executive Summary

Executive Summary

- - The dataset contains 100 rows and 10 columns with no missing values and no duplicate rows.
- - Two KPI columns are provided: Units_Sold (avg 5,128.71; min 124; max 9,925) and Unit_Price (avg 276.76; min 9.33; max 668.27).
- - Categorical columns available for segmentation: Region, Country, Item_Type, Sales_Channel, Order_Priority. Ship_Date is the date column.
- - The most common Region is Sub-Saharan Africa and the most common Country is The Gambia.
- - The strongest reported correlation is a negative correlation of -0.22 between Order_ID and Units_Sold.

Key Findings

- - Data quality is good at surface level:
- - No missing values and no duplicate rows across 100 records.

- - KPI level summary:
- - Units_Sold shows a wide range (124 to 9,925) around an average of 5,128.71, indicating variability in order sizes.
- - Unit_Price also varies substantially (9.33 to 668.27) around an average of 276.76.
- - Categorical coverage enables segmentation by geography (Region, Country), product (Item_Type), channel (Sales_Channel), and priority (Order_Priority).
- - Location concentration:
- - The most frequently occurring Region is Sub-Saharan Africa; the most frequent Country is The Gambia.
- - Correlation insight:
- - The strongest reported relationship is a negative correlation of -0.22 between Order_ID and Units_Sold. This is a weak negative association and does not imply causation.

Business Recommendations

- - Use the available categorical fields to segment performance:
- - Analyze Units_Sold and Unit_Price by Region, Country, Item_Type, Sales_Channel, and Order_Priority to identify high- and low-performing segments.
- - Investigate outliers and variability:
- - Review individual records near the minimum and maximum values for Units_Sold and Unit_Price to understand whether these are valid large/small orders or data anomalies.
- - Leverage Ship_Date for temporal analysis:
- - Perform time-series or seasonality analysis (orders over time) to detect trends, peaks, or recurring patterns.
- - Treat the Order_ID ↔ Units_Sold correlation cautiously:
- - Verify whether Order_ID encodes time or sequence; if so, consider analyzing temporal effects directly via Ship_Date rather than Order_ID.
- - Build simple dashboards for monitoring:
- - Track average Units_Sold and Unit_Price, plus segmentation filters for Region/Country/Item_Type/Sales_Channel to surface actionable patterns.

Risks

- - Sample size limitation:
- - With only 100 rows, findings may not generalize; statistical power is limited for strong inferences.
- - Correlation vs causation:
- - The observed -0.22 correlation between Order_ID and Units_Sold is weak and does not establish causality.
- - Potential outliers:
- - Wide min-max ranges in Units_Sold and Unit_Price suggest possible outliers that could skew averages.
- - Unknown metadata:
- - The meaning and structure of Order_ID are not specified; it could be an identifier without analytic meaning or a proxy for time.
- - Limited KPI set:
- - Only two KPIs are available; additional financial or margin metrics would improve business decision-making.

Next Steps

- - Validate and enrich data:
- - Confirm the meaning and sequencing of Order_ID and confirm Ship_Date formatting and completeness.
- - If possible, increase sample size or obtain more historical data.
- - Descriptive deep-dive:
- - Calculate medians, quartiles, and standard deviations for Units_Sold and Unit_Price; visualize distributions to detect skew and outliers.
- - Segmented analysis:
- - Cross-tabulate KPIs by Region, Country, Item_Type, Sales_Channel, and Order_Priority to find

high-value segments.

- - Time-series analysis:
- - Use Ship_Date to analyze trends, seasonality, and potential temporal drivers of Units_Sold and Unit_Price.
- - Follow-up correlation and causality checks:
- - Re-run correlation analysis using relevant numeric fields and consider regression models only after ensuring adequate sample size and variable meaning.

4. Forecast Analysis

5. Conclusion

This report was automatically generated using DataSage AI. The insights presented are based on statistical analysis, machine learning, and AI-generated business recommendations to support informed decision-making.